

HÜSEYİN YURTSEVEN

Senior Engineering Leader | Machine Learning & Personalization

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EXECUTIVE SUMMARY

Accomplished engineering executive with 20+ years of software development experience and 10+ years leading high-performance teams at scale. Recognized leader in AI/ML and Data Science organizations, with proven expertise building and scaling cross-functional teams of Applied Scientists, Machine Learning Engineers, and Data Scientists. Track record of delivering measurable business impact through ML-powered personalization, recommendation systems, and data-driven decision-making at companies including Amazon, Alibaba Group, and Delivery Hero.

Core Competencies: - AI/ML & Data Science Leadership - Cross-Functional Team Management (Applied Science, ML Engineering, Software Development) - Machine Learning Systems at Scale (Personalization, Ranking, Recommendation) - Distributed Systems & Microservices Architecture - Organizational Design & Team Topologies - Product Development & Strategic Planning

PROFESSIONAL EXPERIENCE

Amazon / Amazon Music

Senior Software Development Manager, Amazon Music Machine Learning *January 2023 - Present*
/ Berlin, Germany

Lead cross-functional teams of Applied Scientists, Machine Learning Engineers, and Software Development Engineers within Amazon Music Personalization organization, driving customer experience optimization and algorithmic innovation.

Key Responsibilities & Achievements: - Direct leadership of cross-functional teams across multiple disciplines (Applied Science, ML Engineering, Software Development) focused on customer understanding, ranking optimization, and experience orchestration - Drive strategic technical initiatives in personalization algorithms, re-ranking systems, and customer behavior modeling - Architect and implement scalable ML systems serving millions of Amazon Music customers globally - Collaborate with product, science, and engineering leadership to define roadmap and technical strategy - Establish engineering excellence practices and mentor senior technical talent

Technical Focus Areas: - Customer Understanding & Behavioral Modeling - Ranking & Re-ranking Algorithms - Experience Optimization & Orchestration - ML Infrastructure & Scalability

Trendyol / Alibaba Group

Head of Engineering *December 2021 - January 2023 / Istanbul, Turkey*

Led engineering organization for local commerce (Food and Groceries Delivery) vertical, managing managers of managers and overseeing 17+ teams across multiple domains including logistics, assortment, and store operations.

Key Responsibilities & Achievements: - Established organizational structure and team topologies for 17+ engineering teams, optimizing for autonomy and delivery velocity - Served on tribe leadership group, shaping company-wide technical strategy and architectural decisions - Partnered with data engineering and data science teams to build data-driven decision-making capabilities - Ensured consistent delivery of

high-quality applications across logistics, assortment, and store operations domains - Implemented resource allocation framework ensuring predictable delivery across multiple concurrent projects

Organizational Impact: - Introduced OKR (Objectives and Key Results) framework across engineering organization - Established Executive Action Team for rapid decision-making on critical technical initiatives - Implemented RFC/ADR process, significantly improving architectural decision quality and documentation - Transformed organization to outcome-driven, empowered team model - Enhanced cross-team collaboration and reduced delivery cycle times

Technologies: Python, FastAPI, Golang, gRPC, React, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, PostgreSQL, Elasticsearch, Nginx, Drone CI/CD, AWS, GCP

Delivery Hero SE

Director, Software Engineering *July 2020 - December 2021 | Berlin, Germany*

Led Q-Commerce engineering tribe, managing managers of managers, senior data science managers, and engineering teams. Drove technical strategy and organizational excellence for rapid-growth business unit.

Key Responsibilities & Achievements: - Directed engineering organization including platform teams, data science teams, and product engineering teams - Served on tribe leadership committee, establishing technical vision and engineering standards - Architected and oversaw delivery of multiple mission-critical applications and platform services - Designed team topologies optimized for business domain alignment and technical dependencies - Managed technical interviews and performance management for senior technical talent

Technical & Business Impact: - Launched comprehensive Product Information Management (PIM) system, improving catalog quality and operational efficiency - Redesigned team structure resulting in measurable improvements in delivery velocity and quality metrics - Established metrics-driven culture with data-informed decision-making frameworks - Implemented OKR methodology across tribe, aligning engineering efforts with business outcomes - Developed multi-criteria ranking solution addressing data discrepancy challenges and improving search relevance

Technologies: Python, FastAPI, Golang, gRPC, React, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, PostgreSQL, Elasticsearch, Varnish, Nginx, Drone CI/CD, TDD, AWS, GCP, Pub/Sub

Kariyer.net

Chief Technology Officer *February 2020 - July 2020 | Istanbul, Turkey*

Led technology organization of Turkey's premier job board and ATS platforms (Kariyer.net and İşin Olsun), managing 80+ engineers, data scientists, DevOps engineers, and QA professionals.

Key Responsibilities & Achievements: - Directed all engineering, R&D, DevOps, and QA functions across multiple product lines - Established software architecture standards and scalability principles for high-traffic platforms - Implemented OKR and Agile methodologies (Scrum, Kanban) across engineering organization - Led recruitment, training, and performance management for technical teams - Ensured predictable delivery of features and applications through resource planning and allocation

Technologies: .NET, .NET Core, Python, Sanic, React, Next.js, Vue.js, Nuxt.js, JWT, scikit-learn, TensorFlow, PyTorch, Microservices, MongoDB, SQL Server, PostgreSQL, Elasticsearch, Varnish, Nginx, Jenkins CI/CD, TDD, Azure

BluTV

Chief Technology Officer *April 2018 - February 2020 / Istanbul, Turkey*

Led technology vision and execution for Turkey's leading OTT streaming platform, managing 40+ engineers, data scientists, product owners, DevOps engineers, and QA professionals.

Key Responsibilities & Achievements: - Directed end-to-end technology development for comprehensive OTT platform serving millions of users - Built in-house capabilities across video delivery, DRM, recommendation systems, and content management - Established engineering culture focused on quality, scalability, and innovation - Managed cross-functional collaboration between engineering, product, and business stakeholders - Implemented Agile methodologies and modern software development practices

Technical Innovations: - Developed proprietary video streaming infrastructure supporting HLS, MPEG-DASH, and Smooth Streaming - Implemented multi-DRM solution (PlayReady, FairPlay, Widevine) for content protection - Built ML-powered recommendation and personalization systems - Architected microservices-based platform for scalability and reliability

Technologies: Python, Sanic, Tornado, React, React Native, Next.js, Vue.js, Nuxt.js, JWT, scikit-learn, Microservices, MongoDB, PostgreSQL, Elasticsearch, Varnish, Nginx, HLS, MPEG-DASH, PlayReady, FairPlay, Widevine, Video.js, Shaka Player, ExoPlayer, Jenkins CI/CD, TDD, AWS

Dogan TV

Director of Software Development and Technical Innovation *September 2015 - April 2018 / Istanbul, Turkey*

Led 70+ person engineering organization across multiple media properties and platforms, driving technical innovation and architectural excellence.

Key Responsibilities & Achievements: - Managed managers of managers, establishing multi-level leadership structure - Directed development of headless CMS (Blupoint CMS), eCRM, DRM systems, and custom CDN solutions - Architected and deployed Hadoop-based ML systems for recommendation, churn prediction, and segmentation - Processed and analyzed clickstream and video event data from 40+ platforms - Implemented full DRM license servers (Widevine, FairPlay, PlayReady)

Platform Portfolio: blutv.com.tr, netd.com, kanald.com.tr, cnnturk.com, dsmart.com.tr, radyonom.com, dreamtv.com.tr

Technologies: Python, Tornado, Vue.js, Nuxt.js, JWT, scikit-learn, Microservices, MongoDB, Elasticsearch, Varnish, Nginx, Hadoop, HLS, MPEG-DASH, Smooth Streaming, DRM technologies

Software Development and Technical Innovation Manager *January 2014 - September 2015*

Managed Dogan TV Software Development Team, leading application engineers and web developers for netd.com, kanald.com.tr, and cnnturk.com.

Key Achievements: - Built multi-site flexible CMS system adopted across all Dogan TV properties - Implemented technology stack: MongoDB, Elasticsearch, Redis, ASP.NET MVC, Web API, OData, BackboneJS, AngularJS - Developed Python-based recommendation systems and ML algorithms - Created HLS-capable Flash and HTML5 video players with VAST 3 advertising support

netd.com - Software Development Manager *August 2012 - January 2014*

Managed netd.com software development team, delivering high-quality applications and establishing architectural standards.

cnnturk.com - Software Development Manager *March 2009 - August 2012*

Led cnnturk.com engineering team, managing weekly releases, capacity planning, and cross-functional coordination between business and engineering.

Key Contributions: - Developed mobile applications for iPhone, iPad, Android, Nokia Ovi, and Samsung Bada platforms - Conducted capacity planning ensuring system stability during high-traffic events - Established release engineering processes and quality standards

cnnturk.com - Senior Software Developer *March 2008 - March 2009*

Promoted to management within one year. Developed cnnturk.com CMS system and maintained internet properties.

VERUS BILISIM

Software Development Team Leader *October 2005 - March 2008 / Istanbul, Turkey*

Led development teams delivering diverse applications including congress management systems, shipping/cargo management software, ERP solutions, and CMS platforms.

Technologies: Visual C#, ASP.NET, SQL Server, PostgreSQL, MySQL, SQL Server Reporting Services, FastReport, Crystal Reports

HYBK BILISIM

Web Developer *January 2002 - June 2005 / Istanbul, Turkey*

Developed web applications and solutions during academic studies.

PUBLICATIONS

A Unified Recommendation Model for Features Summarization Amazon Science <https://www.amazon.science/publications/a-unified-recommendation-model-for-features-summarization>

TECHNICAL SKILLS

Leadership & Management: Coaching, Mentoring, Team Building, Organizational Design, Cross-Functional Leadership, Strategic Planning, OKR Implementation, Agile/Scrum/Kanban Methodologies

Architecture & Systems: Distributed Systems, Microservices Architecture, RESTful APIs, Event-Driven Architecture, Scalability Design, System Design, Cloud Architecture (AWS, GCP, Azure)

Programming & Frameworks: Python (FastAPI, Sanic, Tornado, Django, scikit-learn, TensorFlow, PyTorch), Golang, Java (Spring Boot), JavaScript/TypeScript, Node.js, React, Next.js, Vue.js, Nuxt.js, C#, .NET, ASP.NET MVC

Data & ML: Machine Learning, Recommendation Systems, Personalization, Data Mining, Deep Learning, MLOps

Databases & Storage: MongoDB, PostgreSQL, MySQL, SQL Server, Elasticsearch, Redis, HBase, CouchDB

Infrastructure & DevOps: Docker, Kubernetes, Jenkins, Drone CI, Spinnaker, CI/CD Pipelines, Nginx, Varnish, Apache

Video & Streaming: HLS, MPEG-DASH, Smooth Streaming, DRM (PlayReady, FairPlay, Widevine), Video.js, Shaka Player, ExoPlayer

EDUCATION

Bachelor of Science in Physics Istanbul University | Istanbul, Turkey | 2000 - 2005 *Graduation Thesis: Chaos Theory / Predicting Weather Conditions*

PROFESSIONAL CERTIFICATIONS

Dynamic Adaptive Streaming Over HTTP Using MPEG-DASH Perpetual Solutions | 2014

Project Management and PMP Preparation Yildiz Technical University | 2012

C-SAP Secure Software Development Lostar | 2009

TRAININGS

- AWS Solution Architect (2018)
 - Deep Learning (2018)
 - Hadoop (2018)
 - Blockchain Programming (2018)
 - Product Management (2017)
 - Docker / Containers (2017)
 - Kubernetes (2017)
 - Microservice Fundamentals (2017)
 - Machine Learning (2013)
 - iPad and iPhone Application Development (2011)
 - iPhone Application Development (2010)
 - CMMI (2007)
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PORTFOLIO PROJECTS

GUI for CNC Machines (PLC Controller, CAD/CAM)

2013 - Present

Designed and implemented Visual C# user interfaces for CNC machine control systems for Schtec Machine.

Technologies: Visual C#, SQLite, TwinCAT, CAD/CAM

Multi-Language eCommerce Platform

2009 - Present

Architected and built complete eCommerce framework and platform (www.shoetekfiyat.com).

Technologies: ASP.NET MVC (C#), SQL Server, NHibernate, SQL Server Reporting Services, Bootstrap, LESS, jQuery

Pronet iPhone Panic Button Application

2013

Developed native iOS safety application (Awarded Altın Örümcek - Golden Spider Award).

Technologies: Xcode, Objective-C

Online Poker Hand Replayer

2012

Built interactive hand replay visualization for online poker platforms.

Technologies: ActionScript 3

Microsoft CRM and Navision Add-ons for Windows Mobile

2009

Developed mobile extensions for Microsoft CRM and Navision ERP systems.

Technologies: Visual C#

PROFESSIONAL READING

Selected influential books on leadership, engineering management, and technical excellence:

- Accelerate - Nicole Forsgren PhD
 - Team Topologies - Matthew Skelton and Manuel Pais
 - Multipliers - Liz Wiseman
 - Radical Candor - Kim Scott
 - Effective Executive - Peter Drucker
 - Sprint - Jake Knapp
 - Agile Testing - Janet Gregory
 - The Scrum Fieldbook - Jeff Sutherland
 - Objectives and Key Results - Paul R. Niven
 - Measure What Matters - John Doerr
 - The Manager's Path - Camille Fournier
 - Crucial Conversations - Kerry Patterson et al.
 - Data Mesh - Zhamak Dehghani
 - Collective Intelligence - Toby Segaran
 - Machine Learning: An Algorithmic Perspective - Stephen Marsland
 - Data Mining: Practical Machine Learning Tools and Techniques - Ian H. Witten, Eibe Frank, Mark A. Hall
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ADDITIONAL INFORMATION

Languages: Turkish (Native), English (Fluent) **Location:** Berlin, Germany **Work Authorization:** Permanent Residency

More about my leadership philosophy and working style: <https://hyurtseven81.github.io/>